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Diterpenoid tanshinones inhibit gastric cancer angiogenesis through the PI3K/Akt/mTOR signaling pathway

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ABSTRACT

Ethnopharmacological relevance: *Salvia miltiorrhiza* Bunge is a kind of Chinese herbal medicine known for activating blood circulation and removing blood stasis, with the effect of cooling blood and eliminating carbuncles, and has been proven to have the effect of treating tumors. However, the inhibitory effect of *Salvia miltiorrhiza* Bunge extracts (Diterpenoid tanshinones) on tumors by inhibiting angiogenesis has not been studied in detail. **Aim of the study:** This study aimed to investigate the anti-gastric cancer effect of diterpenoid tanshinones (DT) on angiogenesis, including the therapeutic effects and pathways.

Materials and methods: This experiment utilized network pharmacology was used to identify relevant targets and pathways of *Salvia miltiorrhiza* Bunge-related components in the treatment of gastric cancer. The effects of DT on the proliferation and migration of human gastric cancer cell line SGC-7901 and human umbilical vein endothelial cell line HUVECs were evaluated, and changes in the expression of angiogenesis-related factors were measured. In vivo, experiments were conducted on nude mice to determine tumor activity, size, immunohistochemistry, and related proteins.

Results: The findings showed that DT could inhibit the development of gastric cancer by suppressing the proliferation of gastric cancer cells, inducing apoptosis, and inhibiting invasion and metastasis. In addition, the content of angiogenesis-related factors and proteins was significantly altered in DT-affected cells and animals.

Conclusions: Results suggest that DT has potential as a therapeutic agent for the treatment of gastric cancer, as it can inhibit tumor growth and angiogenesis. It was also found that DT may affect the expression of the angiogenic factor VEGF through the PI3K/Akt/mTOR pathway, leading to the regulation of tumor angiogenesis. This study provides a new approach to the development of anti-tumor agents and has significant theoretical and clinical implications for the treatment of gastric cancer.

1. Introduction

Gastric cancer is a major global health issue with over 1 million new cases and 760,000 deaths reported in 2020 (equivalent to 1 in every 13

deaths in the world), making it the fifth most common cancer and the fourth leading cause of death worldwide (Sung et al., 2021). Although chemotherapy, targeted therapy, immunotherapy, and other systemic treatments have been used to treat gastric cancer, they often come with

Abbreviations: DT, Diterpenoid Tanshinones; DDP, Cisplatin; TCM, Traditional Chinese Medicine; TBST, Tris-buffered saline Tween-20.

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high toxicity, easy recurrence, and poor quality of life for patients (Guan et al., 2023). Therefore, there is a pressing need to develop a safe and effective treatment for gastric cancer.

Tumor angiogenesis plays a crucial role in tumorigenesis and development. It was confirmed that tumor angiogenesis is associated with tumor growth, metastasis, and prognosis (Parmar and Apte, 2021). Consequently, inhibition of tumor angiogenesis has the potential to prevent tumor growth and metastasis (Kanugula et al., 2021). Traditional Chinese medicines with blood-activating effects are effective in treating cancer (Zhang et al., 2021a). *Salvia miltiorrhiza* Bunge (the plant name has been checked with <http://www.theplantlist.org>), named Danshen in China, is a Chinese herbal medicine that has been proven by modern medicine to promote blood circulation and remove blood stasis, mainly used for the prevention and treatment of cardiovascular and cerebrovascular diseases (Wei et al., 2023; Chen et al., 2023). There is no definition of “malignant tumor” in ancient TCM theory, but modern medicine categorizes them into diseases such as “Zhengjia” or “Zhengshi” based on their clinical manifestations. For these types of diseases, *Salvia miltiorrhiza* Bunge has good corresponding therapeutic effects. It has been reported that *Salvia miltiorrhiza* activates downstream eIF4E-binding protein 1 and p70 ribosomal protein S6 kinase through activation of mTOR and initiates vascular endothelial growth factor (VEGF) synthesis (Teng et al., 2020). *Salvia miltiorrhiza* Bunge also contains various chemical components with anti-tumor biological activity. These activities involve multiple mechanisms, including inhibiting cell proliferation and inducing cell apoptosis (Guo et al., 2020).

We found that the main active ingredients in *Salvia miltiorrhiza* Bunge are a series of compounds including Tanshinone I, Tanshinone IIA, and Tanshinone IIB in our previous research (Supplementary Table 1). These active substances play a key role in tumor suppression.

For example, Studies have shown that Tan II induced differentiation of various tumor cells and inhibited cell proliferation, and promote apoptosis (Zu et al., 2018). Moreover, the combination of Tan II and imatinib resulted in lower levels of p-PI3K, p-AKT and p-mTOR in cells and tissues, it also enhanced the effect of imatinib on the biological properties of acute lymphoblastic leukemia cells (Teng et al., 2020). PI3K/AKT/mTOR signaling pathway is the main signaling pathway involved in cell proliferation, apoptosis, and metastasis, and the activation of the PI3K/AKT signaling pathway can promote VEGF synthesis. Multiple studies have shown that Tan IIA may act as a PI3K/AKT/mTOR inhibitor with the ability to inhibit cell proliferation. It significantly reduced the expression of phosphorylated proteins in PI3K, AKT, and mTOR in vivo, and can also inhibit tumor growth by downregulating VEGF (Morin and Canaud, 2021; Di et al., 2017; Liu et al., 2022a). In addition, tanshinones IIA prevented angiogenesis in colorectal cancer by dose-dependently reducing HUVEC proliferation, tube formation and metastasis by decreasing the expression of HIF-1 α , VEGF and bFGF in HCT-116 cells (Zhou et al., 2020a). To further elucidate the role of DT in inhibiting tumour growth and angiogenesis, and its potential as a therapeutic agent for gastric cancer, we performed HPLC assays to obtain a group of compounds with the higher and most potent content, and collectively refer to this group of compounds as Diterpenoid Tanshinones (DT). However, the inhibitory effect of Diterpenoid tanshinones on tumors by inhibiting angiogenesis has not been studied in detail.

Network pharmacology has emerged as a valuable tool for traditional Chinese medicine (TCM) research, facilitating the screening of TCM components, target prediction, and development of new indications for TCM formulations (Zhao et al., 2023).

In this study, the active compounds, potential protein targets, and

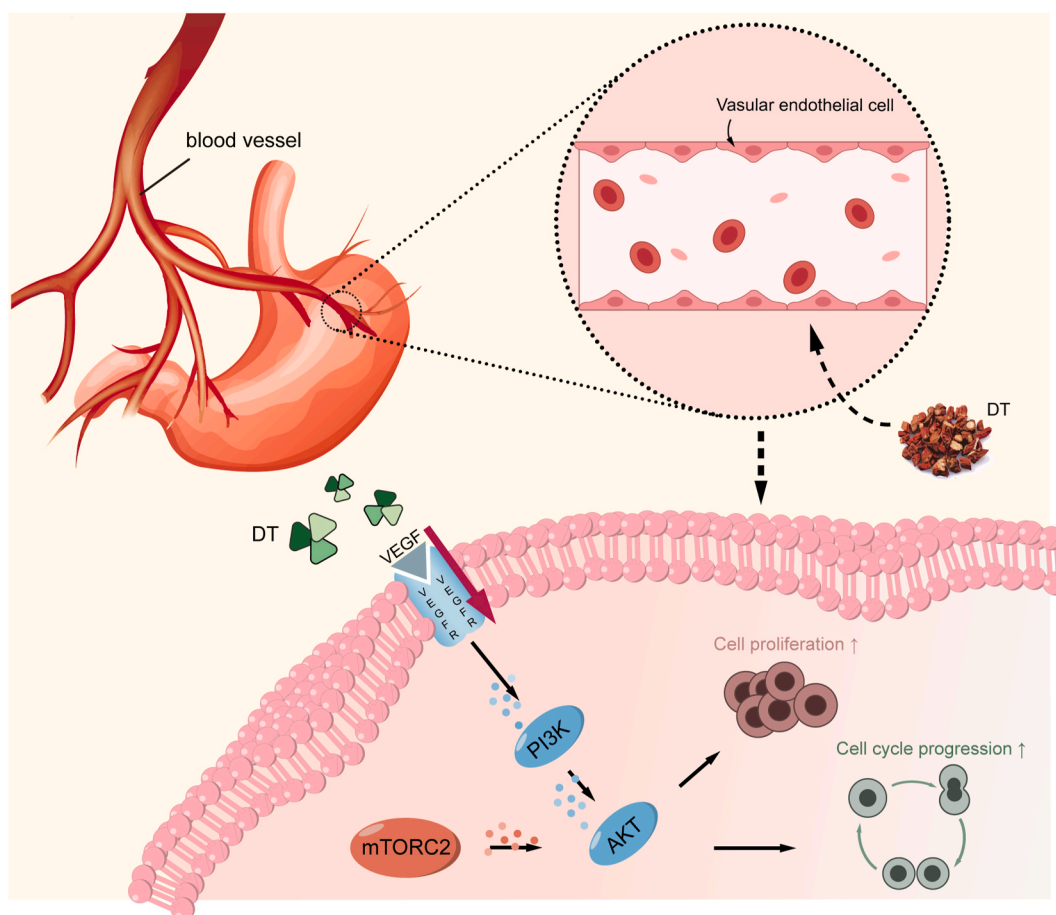


Fig. 1. Experimental mechanisms.

pathways of Diterpenoid tanshinones were predicted by applying a network pharmacology approach. In addition, in vitro and in vivo experiments were performed to verify the potential mechanism of Diterpenoid tanshinones in gastric cancer treatment by inhibiting angiogenesis (Graphical abstract and Fig. 1).

2. Materials and methods

2.1. Network pharmacology-based analysis

The active ingredients and their corresponding targets of *Salvia miltiorrhiza* Bunge were screened from TCMSP (<https://tcmsp.com/tcmssp.php/>) based on oral bioavailability ($OB \geq 30\%$) and drug similarity ($DL \geq 0.18$) criteria. Gastric cancer-related human genes were collected from the GEO DataSets database in NCBI (<https://www.ncbi.nlm.nih.gov/>). From the ingredient-target database using Perl, and the intersection of the two was taken to draw a Venn diagram. The active ingredient-common target network was constructed using Cytoscape 3.7.2 software. Protein interaction (PPI) network map was drawn by String database. GO and KEGG enrichment analysis was performed using R language to clarify the role of target proteins in gene function.

2.2. Experimental drug

Cisplatin (DDP) was purchased from Silojin Biotechnology Co., Ltd., Hangzhou, and it was converted to 0.25 mg/ml.

Diterpenoid tanshinones, with a final mass concentration of 100 mg/ml, were prepared using the "ethanol-petroleum ether" extraction method and provided by the Pharmacy Laboratory of Zhejiang University of Traditional Chinese Medicine.

2.3. Experimental cell

Human gastric cancer cell line SGC-7901 and human umbilical vein endothelial cell line HUVECs were purchased from the Shanghai Chinese Academy of Sciences Type Culture Collection Cell Bank. SGC-7901-luc cells was transfected with HBLV-luc-puro lentivirus (3×10^8 TU/ml, Hanhen Biologicals, Inc.). SGC-7901 cells were seeded in 24-well plates at a concentration of 1×10^5 cells/well and growing to the confluence of 30%–40%, then add lentivirus to make $MOI = 15$. 6 μ g/ml polybrene was added as a co-transfection agent to each well. 24h after infection, the virus-containing culture medium was replaced with a fresh medium. SGC-7901-luc cell lines were screened with 2 μ g/ml puromycin 72h after infection and transfection efficiency was detected by chemiluminescence detector.

SGC-7901 or SGC-7901-luc cells were cultured in RPMI-1640 (GIBCO, USA) containing 10% FBS (GIBCO, USA), HUVECs were cultured in DMEM (containing 10% FBS), and double antibody 100U/ml penicillin, 100 μ g/ml chain Mycin (GENOM BIO Co., Ltd., Hangzhou), and placed in a 5% CO₂, 37 °C incubator for cultivation.

2.4. Animals

BALB/c-nu nude mice, 3–5 weeks old, male, 18 ± 3 g, were purchased from the Animal Experiment Center of Zhejiang University of Traditional Chinese Medicine (entrusted to Shrek Co., Ltd., Shanghai for supply) and raised in SPF grade of the Animal Experiment Center of Zhejiang University of Traditional Chinese Medicine Barrier animal room, animal operation standards are by the requirements of the center.

2.5. Cell viability assay

SGC-7901 and HUVECs cells (1.5×10^4 cells/well) were seeded in a 96-well plate. After the cells were in the logarithmic growth phase, added fresh drug-containing medium. Add different concentrations of DT to the cells. After 24h of culture, add 10 μ l Cell Counting Kit-8

(DOJINDO, Japan) to each well, and measure the OD value at 450 nm. Calculate the mean OD value of each group of replicate wells calculating cell viability.

2.6. Cell cycle analysis

SGC-7901 and HUVECs cells (5×10^5 cells/well) were seeded in a 6-well plate. After overnight incubation, different concentrations of DT solution or cisplatin were added. After 24 h, harvest the cells and wash them with cold PBS. Fix the cells overnight with 70% ethanol at 20 °C and stain with propidium iodide (PI) in the dark at 37 °C for 30 min. Then, flow cytometry was used to detect the distribution of cell cycles.

2.7. Scratch assay

SGC-7901 and HUVECs cells (7×10^6 cells/well) were seeded in a 6-well plate. After culturing for 24 h, a wound was created in the cell plane using the tip of a 200 μ l pipette and all floating cells were gently removed by PBS, and then add different concentrations of DT. Observe and take photos at the same location at 0, 6, 24, 48, and 96 h, and compare the distance of cell migration.

2.8. Tube formation assay

SGC-7901 and HUVECs cells (3.5×10^4 cells/well) were added to 96 well plates containing Matrigel glue after being treated with different concentrations of DT and DDP for 24h. Take pictures after the tube is formed.

2.9. Migration and invasion ability

After a 12-h period of cellular starvation, SGC-7901 cells were suspended in serum-free medium with varying concentrations of DT and cisplatin at 5×10^5 cells/ml. Subsequently, 200 μ l of the cell was introduced into the upper chamber, while 750 μ l of serum-containing medium was placed in the lower chamber. For the invasion experiment, a cold serum-free culture medium was combined with Matrigel glue at an 8:1 ratio, and 100 μ l of this mixture was added to each well in the upper chamber of the transwell. Following a 24-h incubation period, the upper chamber underwent fixation with 4% formaldehyde for 20 min. Then the cells were stained with crystal violet for 20 min. Wipe the cells on the upper surface of the filter membrane with a cotton swab. Migration towards the bottom was observed under an optical microscope in accordance with standard scientific publication criteria.

2.10. RT-qPCR analysis

Cells were homogenized and placed into TRI reagent for RNA extraction, and the concentration of RNA was determined using NanoDrop® ND-2000. The RT-qPCR reaction was performed after reverse transcription of RNA (1 μ g) into cDNA according to the instructions of the Real-time PCR kit (Takara, Dalian). Cycling conditions were 95 °C for 30s, followed by 40 cycles at 95 °C for 5 s, 60 °C for 30 s. This was followed by a final extension of 15s at 95 °C, 30s at 60 °C, and 15s at 95 °C. All the primers used in this study are listed in [Supplementary Table 2](#).

2.11. Western blot analysis

Proteins were extracted from cells or tissues by lysis buffer (Applygen Co., Ltd., Beijing) on ice for 30mins. Lysates were centrifuged, and supernatants were collected. Denaturing the protein at 100 °C and total protein concentrations (mg/ml) were determined by the BCA reagent assay (Applygen Co., Ltd., Beijing). Subsequently, SDS-PAGE electrophoresis, membrane transfer, and closure were performed for 2h. After soaking overnight with the primary antibody, rinse TBST three times,

then soak and incubate with the secondary antibody for 2 h, and rinse TBST three times, each time for 10 min. Two reagents from the ECL luminescence kit, A and B, were mixed in equal volumes in centrifuge tubes, and the ECL mixture was added dropwise on PVDF membrane, protected from light for 3–5 min, developed and photographed on the instrument, and recorded. The experimental results were recorded. The primary antibodies are following: VEGF (1:1000; 19003-1-AP; Proteintech Group), VEGFR1 (1:600; 13687-1-AP; Proteintech Group), Endostain (1:1000; ab275746 abcam), FGF-2 (1:1000; ab208687; abcam), FGF-7 (1:1000; ab131162; abcam), mTORC2(RICTOR)(1:2000; 27248-1-AP; Proteintech Group), FOXO1(1:500; 18592-1-AP; Proteintech Group), PI3K(1:3000; 60225-1-Ig, Proteintech Group), AKT (1:1000; 10176-2-AP; Proteintech Group), GAPDH (1:5000, 60004-1-Ig; Proteintech Group). The secondary antibodies are goat anti-rabbit (1:10000; 111-035-003; Jackson) and Goat anti-mouse (1:10000; 115-035-003; Jackson).

2.12. Establishment of orthotopic transplantation tumor model

In the axillary region, mice received 1×10^7 subcutaneous implants of SGC-7901-luc cells. After 14 days, extracted the tissues of the tumor and cut into small pieces of about 1 mm^3 in normal saline and implanted into the gastric wall of the mouse by surgery. One week later, tumor growth was analyzed by an in vivo mouse imaging system, regrouped based on fluorescence intensity. Mice were divided into a control group, DT Low dose 30 mg/kg, DT Medium dose 60 mg/kg, DT High dose 120 mg/kg, and cisplatin group. The control group was gavaged with 0.2 ml of CMC-Na solution daily and injected intraperitoneally with 0.2 ml sterile saline every 3 days. The DT group was gavaged with 0.2 ml of different concentrations of DT daily and injected intraperitoneally with 0.2 ml sterile saline every 3 days. The cisplatin group was gavaged with 0.2 ml of CMC-Na solution daily and injected intraperitoneally with 0.2 ml of DDP every 3 days. Weights and intravital imaging were performed every 7 days. Four weeks after administration, the mice were sacrificed by cervical dislocation, and the tumor stomach and other organs were removed.

The number of mice in each group at the end of the experiment was: 9 mice in the control group, DT Low dose, DT Medium dose, and cisplatin group, 7 mice in DT High dose.

2.13. Immunohistochemical analysis

The 3- μm -thick deparaffinized tissue sections were subjected to heat-based antigen retrieval with pH = 6.0 citric acid buffer. The sections were then incubated with 1:200 primary antibody at 37 °C for 1h and washed three times with PBS for 5mins each time. The sections were then incubated with secondary antibody at 37 °C for 1h and washed three times with PBS for 5mins each time. Then the nuclei were stained with DAB and hematoxylin. Finally, the slices were sealed with neutral gum. The antibody used in this step is the same as in 2.2.10.

2.14. Statistical analysis

SPSS 22.0 was used to perform a one-way ANOVA analysis on the data, and the measurement data were expressed as mean $\pm$ standard deviation ($\bar{X} \pm s$). When the variance was homogeneous, the LSD-t test was selected; when the variance was unequal, the Dunnett-t test was selected. $p < 0.05$ was considered significant.

3. Result

3.1. Analysis based on network pharmacology

3.1.1. Data preparation

65 quinone active components, including Dehydrotanshinone II A, Poriferasterol, isoimperatorin, etc., were screened from the TCMSP

database for *Salvia miltiorrhiza* Bunge based on the conditions of oral bioavailability (OB) $\geq 30\%$ and drug similarity (DL) ≥ 0.18 (Table 1).

Subsequently, 826 targets corresponding to these active components were obtained. Among them, 327 targets were highly relevant to gastric cancer, including 44 common drug-disease targets (Fig. 2A).

3.1.2. Network construction

Network Construction: The salvia diterpene quinone regulatory network was constructed using Cytospace 3.7.2 software, resulting in a network with 98 nodes and 176 interaction lines. Triangle nodes in the inner circle represent the intersection genes of active components of *Salvia miltiorrhiza* Bunge and gastric cancer targets, while circles in the outer circle represent active components that can act on the intersecting genes (Fig. 2B).

The 44 common targets were imported into the String platform, and protein interaction results were obtained (Fig. 2C). Topological analysis of the PPI network was performed, and key targets were identified using the CytoNCA function package (Fig. 2D). Topological parameter analysis was performed on the top 15 targets (Table 2).

3.1.3. Enrichment analysis of DT's effects on gastric cancer cells

The analysis results of the biological process (BP), cell composition (CC), and molecular function (MF) involved in the core target were obtained in GO analysis. The core target has the highest significance in the response to steroid hormones in biological processes (BP) (Fig. 2E) and the response of cells to chemical stress, followed by metal ion response, drug response, and so on. In terms of cellular composition (CC) (Fig. 2F), the above core targets are most prominent in transcriptional regulatory complexes, followed by cyclin-dependent protein kinase holoenzyme complexes, and so on. In terms of molecular functions (MF) (Fig. 2G), the binding of DNA-binding transcription factors is the most significant, followed by the binding of ubiquitin-like protein ligase. Accordingly, DT may focus on the above biological functions involved in the treatment of gastric cancer.

The KEGG enrichment analysis showed that the core target was in the PI3K-Akt signaling pathway, Kaposi sarcoma-related herpesvirus infection is significant, indicating that DT may achieve the purpose of treating diseases by acting on these signaling pathways (Fig. 2H). Based on this, a KEGG relational network is constructed, the outer circle is the related pathway, the inner circle is the core gene, and its graph size. It is proportional to the number of associated pathways (Fig. 2I).

3.2. Experimental validation

3.2.1. Growth curve of SGC-7901, HUVECs cells

SGC-7901 and HUVECs cells showed different growth patterns. SGC-7901 cells had a uniform proliferation rate with a slow rise, while HUVECs cells had a relatively flat proliferation curve in the first 72 h and accelerated growth thereafter. (Fig. 3A).

3.2.2. The effect of drugs on the proliferation of cells

The lethality of both SGC-7901 cells and HUVECs cells at DT concentrations showed a certain concentration dependence, that is, the higher the concentration, the lower the survival rate, and the mean IC50 of SGC-7901 cells was calculated to be $(11.31 \pm 1.19) \mu\text{g/ml}$ (Fig. 3B). This study focuses on the inhibition of angiogenesis on tumors. Excessively high concentrations of DT can affect tumor cells themselves by killing them, which has an impact on the results of the study. Therefore, the subsequent experimental concentrations of DT were determined as 6 $\mu\text{g/ml}$, 7 $\mu\text{g/ml}$, and 8 $\mu\text{g/ml}$. The mean IC50 of HUVECs was calculated to be $(22.84 \pm 0.56) \mu\text{g/ml}$ (Fig. 3C). Follow-up experimental concentrations were determined to be: 17 $\mu\text{g/ml}$, 20 $\mu\text{g/ml}$, and 23 $\mu\text{g/ml}$.

The lethality of both SGC-7901 cells and HUVECs cells at different DDP concentrations showed a certain concentration dependence. The calculated IC50 mean of SGC-7901 cells is $(1.79 \pm 0.02) \mu\text{g/ml}$ (Fig. 3D). The concentration of the follow-up experiment was 1.3 $\mu\text{g/ml}$.

Table 1
Active ingredients of diterpenoid tanshinones.

Source	Ingredients	OB	DL
Salvia miltiorrhiza	1,2,5,6-tetrahydrotanshinone	38.75	0.36
Salvia miltiorrhiza	Poriferasterol	43.83	0.76
Salvia miltiorrhiza	poriferast-5-en-3beta-ol	36.91	0.75
Salvia miltiorrhiza	isoimperatorin	45.46	0.23
Salvia miltiorrhiza	sugiol	36.11	0.28
Salvia miltiorrhiza	Dehydrotanshinone II A	43.76	0.4
Salvia miltiorrhiza	Baicalin	40.12	0.75
Salvia miltiorrhiza	digallate	61.85	0.26
Salvia miltiorrhiza	luteolin	36.16	0.25
Salvia miltiorrhiza	α -amyrin	39.51	0.76
Salvia miltiorrhiza	5,6-dihydroxy-7-isopropyl-1,1-dimethyl-2,3-dihydrophenanthren-4-one	33.77	0.29
Salvia miltiorrhiza	2-isopropyl-8-methylphenanthrene-3,4-dione	40.86	0.23
Salvia miltiorrhiza	3 α -hydroxytanshinonella	44.93	0.44
Salvia miltiorrhiza	(E)-3-[2-(3,4-dihydroxyphenyl)-7-hydroxy-benzofuran-4-yl] acrylic acid	48.24	0.31
Salvia miltiorrhiza	4-methylenemiltirone	34.35	0.23
Salvia miltiorrhiza	2-(4-hydroxy-3-methoxyphenyl)-5-(3-hydroxypropyl)-7-methoxy-3-benzofurancarboxaldehyde	62.78	0.4
Salvia miltiorrhiza	6-o-syringyl-8-o-acetyl shanzhiside methyl ester	46.69	0.71
Salvia miltiorrhiza	formyltanshinone	73.44	0.42
Salvia miltiorrhiza	3-beta-Hydroxymethylenetanshinquinone	32.16	0.41
Salvia miltiorrhiza	Methylenetanshinquinone	37.07	0.36
Salvia miltiorrhiza	Przewalskin a	37.11	0.65
Salvia miltiorrhiza	Przewalskin b	110.32	0.44
Salvia miltiorrhiza	Przewaquinone B	62.24	0.41
Salvia miltiorrhiza	Przewaquinone c	55.74	0.4
Salvia miltiorrhiza	(6S,7R)-6,7-dihydroxy-1,6-dimethyl-8,9-dihydro-7H-naphtho[8,7-g] benzofuran-10,11-dione	41.31	0.45
Salvia miltiorrhiza	Przewaquinone f	40.31	0.46
Salvia miltiorrhiza	sclareol	43.67	0.21
Salvia miltiorrhiza	tanshinaldehyde	52.47	0.45
Salvia miltiorrhiza	Danshenol B	57.95	0.56
Salvia miltiorrhiza	Danshenol A	56.97	0.52
Salvia miltiorrhiza	Salvilenone	30.38	0.38
Salvia miltiorrhiza	cryptotanshinone	52.34	0.44
Salvia miltiorrhiza	dan-shexinkum d	38.88	0.55
Salvia miltiorrhiza	danshenspiroketallactone	50.43	0.31
Salvia miltiorrhiza	deoxyneocryptotanshinone	49.4	0.29
Salvia miltiorrhiza	dihydrotanshinlactone	38.68	0.32
Salvia miltiorrhiza	dihydrotanshinoneI	45.04	0.36
Salvia miltiorrhiza	epidanshenspiroketallactone	68.27	0.31
Salvia miltiorrhiza	C09092	36.07	0.25
Salvia miltiorrhiza	isocryptotanshi-none	54.98	0.39
Salvia miltiorrhiza	Isotanshinone II	49.92	0.4
Salvia miltiorrhiza	manool	45.04	0.2
Salvia miltiorrhiza	microstegiol	39.61	0.28
Salvia miltiorrhiza	miltiononeI	49.68	0.32
Salvia miltiorrhiza	miltiononeII	71.03	0.44
Salvia miltiorrhiza	miltipolone	36.56	0.37
Salvia miltiorrhiza	Miltirone	38.76	0.25
Salvia miltiorrhiza	miltironeII	44.95	0.24
Salvia miltiorrhiza	neocryptotanshinone ii	39.46	0.23
Salvia miltiorrhiza	neocryptotanshinone	52.49	0.32
Salvia miltiorrhiza	1-methyl-8,9-dihydro-7H-naphtho[5,6-g]benzofuran-6,10,11-trione	34.72	0.37
Salvia miltiorrhiza	prolithospermic acid	64.37	0.31
Salvia miltiorrhiza	(2R)-3-(3,4-dihydroxyphenyl)-2-[(Z)-3-(3,4-dihydroxyphenyl)acryloyl]oxy-propionic acid	109.38	0.35
Salvia miltiorrhiza	(Z)-3-[2-[(E)-2-(3,4-dihydroxyphenyl)vinyl]-3,4-dihydroxy-phenyl]acrylic acid	88.54	0.26
Salvia miltiorrhiza	salvianolic acid g	45.56	0.61
Salvia miltiorrhiza	salvianolic acid j	43.38	0.72
Salvia miltiorrhiza	salvilenoneI	32.43	0.23
Salvia miltiorrhiza	salviolone	31.72	0.24
Salvia miltiorrhiza	NSC 122421	34.49	0.28
Salvia miltiorrhiza	(6S)-6-hydroxy-1-methyl-6-methylol-8,9-dihydro-7H-naphtho[8,7-g]benzofuran-10,11-quinone	75.39	0.46
Salvia miltiorrhiza	Tanshindiol B	42.67	0.45
Salvia miltiorrhiza	Przewaquinone E	42.85	0.45
Salvia miltiorrhiza	tanshinone iia	49.89	0.4
Salvia miltiorrhiza	(6S)-6-(hydroxymethyl)-1,6-dimethyl-8,9-dihydro-7H-naphtho[8,7-g]benzofuran-10,11-dione	65.26	0.45
Salvia miltiorrhiza	tanshinoneVI	45.64	0.3

The calculated IC50 mean of HUVECs cells is (52.23 $\pm$ 23.20) μ g/ml (Fig. 3E). The concentration of the follow-up experiment was 16 μ g/ml.

3.2.3. The effect of DT on the cell cycle

We found that in 0,6,7,8 μ g/ml DT and DDP, 7.89 %, 6.43 %, 6.63 %, 9.71 %, and 9.62 % of SGC-7901 cells were located in the G2/M phase, respectively. The proportions of the cells in the G2/M phase were

significantly higher after 8 μ g/ml DT, and the proportion was lower under the action of other drugs. (Fig. 3F).

We found that in 0,17,20,23 μ g/ml DT and DDP, 9.03 %, 4.51 %, 2.27 %, 10.42 %, and 1.59 % of HUVECs cells were located in the G2/M phase, respectively. The proportions of the cells in the G2/M phase were higher after 23 μ g/ml DT, and the proportion was lower under the action of other drugs. (Fig. 3G).

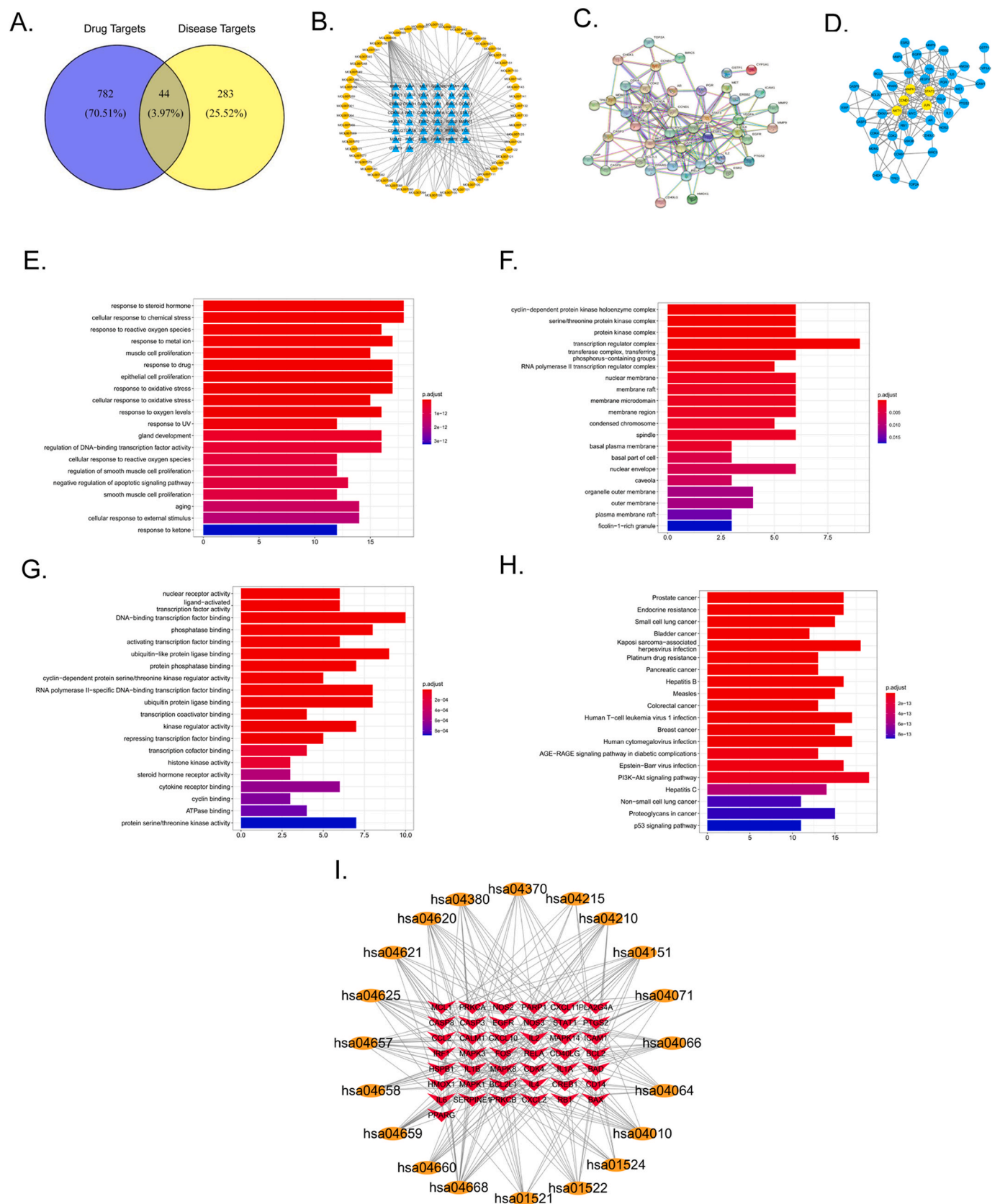


Fig. 2. Network pharmacological study of DT in the treatment of gastric cancer (A) Venn diagram of common drug-disease targets (B) Regulation network diagram of DT (C) Protein interaction network and targets of Diterpenoid tanshinones in the treatment of gastric cancer (D) Identify key targets of Diterpenoid tanshinones in the treatment of gastric cancer (E–G) GO enrichment analysis of Diterpenoid tanshinones (H) Diterpenoid tanshinones KEGG enrichment analysis chart (I) Relationship network.

Table 2

Topological parameter analysis of protein interactions (top 15).

Number	Gene	Degree	BC	CC
1	STAT3	21	373.9	0.2848
2	AKT1	20	314.9	0.2848
3	JUN	17	140.8	0.2774
4	MAPK1	17	171.4	0.2774
5	CCND1	15	129.5	0.2774
6	MYC	14	65.7	0.2739
7	FOS	12	63.0	0.2606
8	IL6	12	56.5	0.2560
9	CDKN1A	11	65.0	0.2671
10	ESR1	11	47.5	0.2575
11	RB1	11	54.1	0.2590
12	RELA	10	35.1	0.2544
13	VEGFA	10	52.6	0.2575
14	CDK2	9	56.2	0.2515
15	EGFR	8	9.9	0.2529

3.2.4. The effect of DT on cell function

In the scratch assay, after treating the cells with different concentrations of DT and DDP, the scratch area decreased over time in all groups of both SGC-7901 cells and HUVECs cells and the higher the DT concentration, the slower the rate of healing (Fig. 3H and J). At the end of the experiment, the blank area was lower in both DT groups than in the DDP group (Fig. 3I and K).

In the tubule formation assay, In the presence of DT, tubule formation in both SGC-7901 cells and HUVECs cells was significantly reduced in the DT medium-dose group and DT high-dose group compared with the control group ($P < 0.05$) and DDP group ($P < 0.05$) (Fig. 4A and B).

In the transwell assay, after treating the cells with different concentrations of DT, the number of permeable cells was significantly lower in the DT group compared to the control group. The number of permeable cells in the DT high-dose group was significantly lower than that in the DDP group. The number of transmembrane cells showed a trend of control group > DT low dose group > DT medium dose group > DDP group > DT high dose group (Fig. 4C and D).

3.2.5. RT-qPCR to detect the effect of DT on the expression of angiogenesis-related genes in SGC-7901 cells

After cells were treated with 3 different concentrations of DT (6, 7, 8 $\mu\text{g/ml}$) and cisplatin (1.3 $\mu\text{g/ml}$), with the increase of DT concentration, the levels of VEGF, VEGFR1, PI3K, AKT, FGF-2, FGF-7, and mTORC2 all showed a decreasing trend, which was negatively correlated with DT concentration. Endostatin and FOXO1 showed a gradually increasing trend, which was positively correlated with DT concentration (Fig. 5A).

3.2.6. Western Blot to detect the effect of DT on the expression of angiogenesis-related proteins in SGC-7901 cells

Cells were treated with different concentrations of DT (6, 7, 8 $\mu\text{g/ml}$) and cisplatin (1.3 $\mu\text{g/ml}$). The protein expression levels of VEGF, VEGFR1, PI3K, AKT, FGF-2, FGF-7, and mTORC2 in cells exhibited a negative correlation with DT concentration, decreasing with the increase of DT concentration. In contrast, Endostatin and FOXO1 showed a positive correlation with DT concentration, exhibiting a gradual increase (Fig. 5B and C).

3.2.7. The effect of DT on nude mice

One week after orthotopic transplantation of gastric cancer, except for 2 nude mice, all tumors could be imaged, and it was considered that the subcutaneous implantation tumor model was successfully constructed.

In the presence of DT, the mean body weight of nude mice was reduced, but there was no significant concentration-dependent trend. (Fig. 6A). Under the effect of DT, the tumor volume of mice gradually decreased (Fig. 6B). At the same time, the fluorescence expression intensity of tumors decreased (Fig. 6C and D) and the tumor weight

decreased (Fig. 6E) with a concentration-dependent trend.

3.2.8. Tumor immunohistochemistry probes the effect of DT on angiogenesis-related antibodies in tumors

Immunohistochemical detection of relevant genes was performed on tumor tissue sections, and brown-stained cells were positive cells. In the presence of DT, the expression of Endostatin was increased and the number of positive cells was positively correlated with DT concentration. the expression of AKT, Ang-2, VEGF, and PCNA was decreased and the number of positive cells was negatively correlated with DT concentration (Fig. 7A).

3.2.9. Western Blot to detect the effect of DT on the expression of angiogenesis-related proteins in tumor

After treating cells with three different concentrations of DT (6, 7, and 8 $\mu\text{g/ml}$) and cisplatin (1.3 $\mu\text{g/ml}$), the protein expression levels of VEGF, VEGFR1, PI3K, AKT, FGF-2, FGF-7, and mTORC2 in tumor tissues showed a decreasing trend with increasing DT concentration, which was negatively correlated with DT concentration. Endostatin and FOXO1 showed a gradually increasing trend, which was positively correlated with DT concentration. (Fig. 7B and C).

4. Discussion

Gastric cancer is a prevalent type of cancer with a high global incidence (Sung et al., 2021; Machlowska et al., 2020). Related studies have shown that the main components of *Salvia miltiorrhiza* Bunge have a wide range of anti-tumor biological activities for diseases including gastric cancer (Lin et al., 2015), and their possible mechanisms of action involve several aspects such as inhibition of cell proliferation, induction of apoptosis, inhibition of tumor invasion and metastasis, inhibition of angiogenesis, and reversal of tumor multidrug resistance (Li et al., 2021; Zhu et al., 2016; Tse et al., 2013; Liu et al., 2021).

Diterpenoid tanshinones (DT), one of the effective components of *Salvia miltiorrhiza* Bunge, have also shown extensive antitumor effects (Zhou et al., 2020b; Li et al., 2022; Sun et al., 2021). DT has good antitumor activity in vitro, with synergistic potentiation and toxicity reduction, which may be related to the mitochondrial pathway and activation of JNK and p38 MAPK signaling pathways (Su et al., 2012; Wang et al., 2019). Tanshinone IIA impedes the invasion and migration of HC8693 cells by inhibiting angiogenesis, while inactivation of the Wnt/ β -catenin axis and reductions in COX-2 and VEGF lead to impaired angiogenesis in tumor cells, thereby affecting tumor cell growth and migration (Ma et al., 2018; Fu et al., 2017).

Since Judah Folkman introduced the concept of tumor angiogenesis, suggesting that inhibition of tumor angiogenesis would stop tumor growth (Folkman, 1971), several studies have confirmed that tumor angiogenesis is closely related to tumor growth, metastasis, and prognosis (Lugano et al., 2020). Therefore, inhibition of tumor angiogenesis has become one of the important pathways for the treatment of cancer. While DT has been shown to have broad antitumor activity, its relationship with tumor angiogenesis has been rarely reported.

We performed a multi-component, multi-target, multi-pathway prediction of *Salvia* involvement in gastric cancer through a network pharmacology study, revealing potential targets and pathways for *Salvia* treatment of gastric cancer. We found that the core targets of *Salvia*-gastric cancer action include Akt and VEGFA, and the potential pathways include the PI3K-Akt signaling pathway.

In this study, we demonstrated that DT inhibits the generation and progression of gastric cancer as well as down-regulates the expression level of Akt, PI3K and mTORC in gastric cancer cells and tumor tissues. Activation of Akt induces different mechanisms of cell survival (Revathidevi and Munirajan, 2019; Jie et al., 2016). Akt inactivates cells by phosphorylating the pro-apoptotic factors Bad and procaspase-9, as well as a family of forkhead factor transcription factors that induce the expression of pro-apoptotic factors such as Fas ligands (Revathidevi and

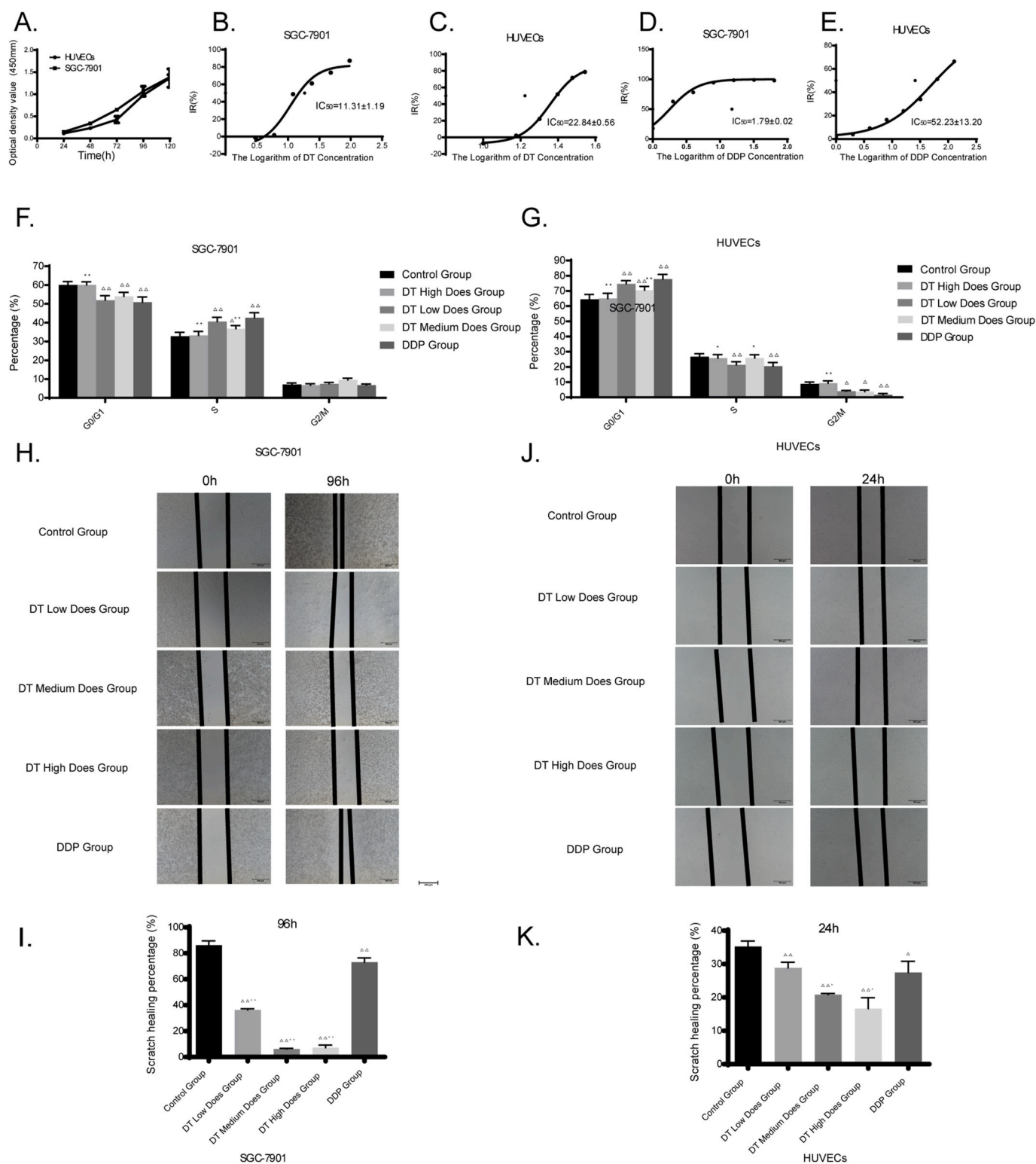


Fig. 3. Cell growth curve, cycle and scratch test (A) Growth curve of SGC-7901, HUVECs cell (B) Growth inhibition rate of SGC-7901 cells by different concentrations of DT intervention (C) The growth inhibition rate of HUVECs cells by different concentrations of DT intervention (D) The growth inhibition rate of SGC-7901 cells by the intervention of different concentrations of cisplatin (E) The growth inhibition rate of HUVECs cells treated with different concentrations of cisplatin (F) Effects of low, medium and high doses of DT on different phases of SGC-7901 cells Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: $\ast p < 0.05$; $\ast\ast p < 0.01$ (G) Effects of low, medium and high doses of DT on different phases of HUVECs Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: $\ast p < 0.05$; $\ast\ast p < 0.01$ (H) Effects of low, medium and high doses of DT at different times on scratching of SGC-7901 cells (I) Effects of low, medium and high doses of DT at different times on the percentage of scratch healing in SGC-7901 cells. Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: $\ast p < 0.05$; $\ast\ast p < 0.01$ (J) Effects of low, medium and high doses of DT at different times on the scratching of HUVECs cells (K) Effects of low, medium and high doses of DT at different times on the percentage of scratch healing in HUVECs cells. Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: $\ast p < 0.05$; $\ast\ast p < 0.01$.

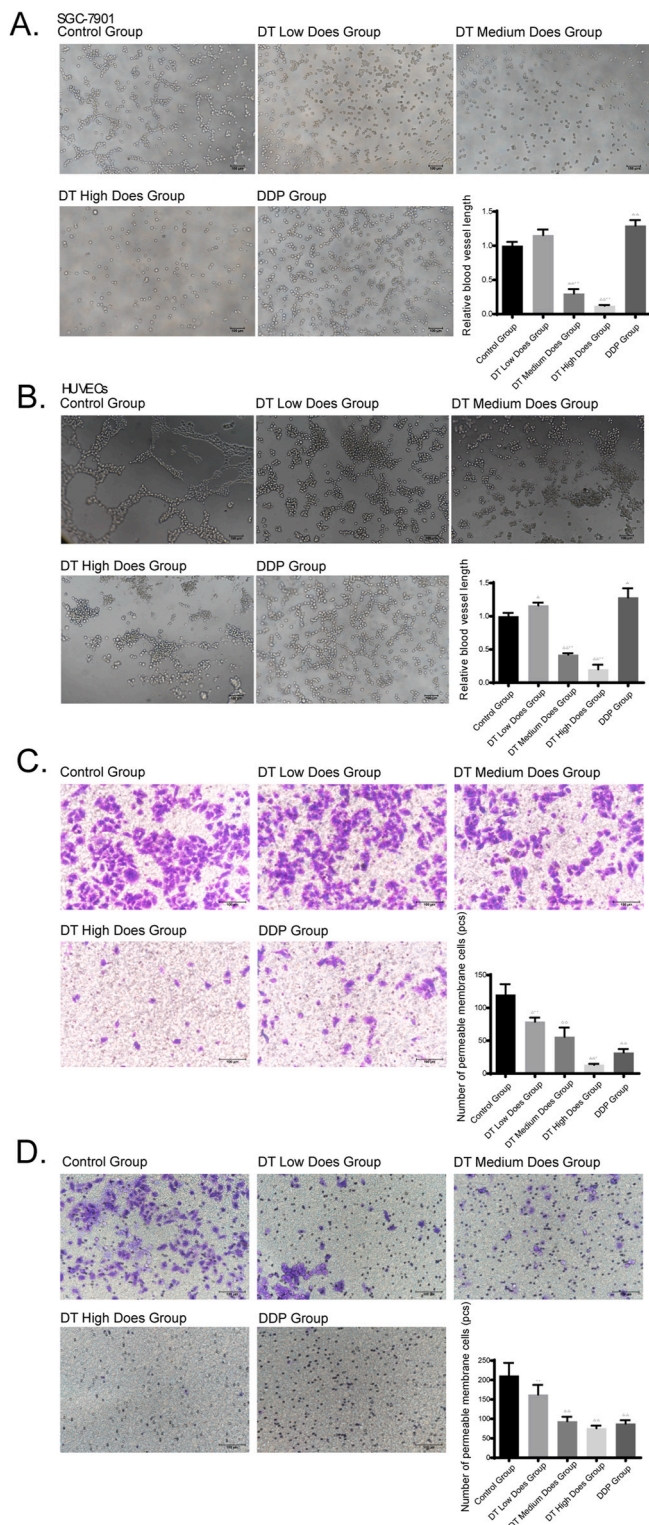


Fig. 4. Tube formation and transwell assay of cells (A) Effects of DT on tube formation in SGC-7901 cells (B) Effects of low, medium and high doses of DT on tube formation in HUVECs (C) Effects of low, medium and high doses of DT on Transwell migration of SGC-7901 cells. Note: Compared with the control group: $\Delta p < 0.05$, $\Delta\Delta p < 0.01$; Compared with the cisplatin group: $*p < 0.05$; $**p < 0.01$ (D) Effects of low, medium and high doses of DT on Transwell invasion of SGC-7901 cells. Note: Compared with the control group: $\Delta p < 0.05$, $\Delta\Delta p < 0.01$; Compared with the cisplatin group: $*p < 0.05$; $**p < 0.01$.

Munirajan, 2019; Wang et al., 2002). The most common pathway in which Akt is involved is the PI3K/Akt/mTOR signaling pathway, which has been shown to be elevated in expression in a variety of cancers, including gastric cancer (Fattahi et al., 2020; Glaviano et al., 2023). Phosphoinositide 3-kinases (PI3Ks) are lipid kinases, which have three subclasses, namely class I, II, and III. PIP3 is created when fully activated PI3K phosphorylates the third carbon of PIP2's inositol head. PIP3 can bind to the N-terminal PH domain of protein kinase B(PKB, Akt), which transfers Akt from the cytoplasm to the cell membrane and activates Akt with the assistance of PDK1 and PDK2 (Liu et al., 2022b). Akt activates mTORC1 directly by phosphorylating mTORC1 at Ser2448, and indirectly by phosphorylating TSC2 (Osaki et al., 2004; Hennessy et al., 2005). The PI3K/Akt/mTOR signaling pathway is crucial for controlling the metabolism, angiogenesis, transcription, translation, and growth of cells (Carnero et al., 2008).

VEGF-A (also commonly referred to as VEGF) can bind to VEGFR2 to activate the RAS-RAF-MAPK-ERK signaling pathway and promote endothelial cell proliferation (Herbert and Stainier, 2011), which is the main regulatory mechanism of angiogenesis. Endothelial cell migration is a key step in angiogenesis and can be triggered by VEGFA. Thus, the expression of VEGF in cells and tissues can reflect the level of angiogenesis. The experiments in this study confirmed that VEGF levels in both gastric cancer cells and tumor tissues tended to decrease under the effect of DT.

Meanwhile, we also measured the expression levels of additional angiogenesis-related factors in cells and tissues. The results showed that the expression levels of VEGFR1, FGF-2 and FGF-7 decreased and the expression of Endostatin and FOXO1 increased in response to DT. VEGFR-1 is one of the receptors of VEGF, and it shows a positive correlation with VEGF expression in tumors (Lacal and Graziani, 2018). Blocking VEGFR-1 can inhibit tumor-related angiogenesis. Endostatin can downregulate VEGF and VEGF receptor (VEGFR) expression to inhibit tumor angiogenesis and block VEGF-C signaling to inhibit tumor lymphatic angiogenesis (Sarbasov et al., 2005). The FGF family acts synergistically with VEGF to promote the growth and maturation of tumor blood vessels. FOXO1 is a transcription factor for forkhead box O (FOXO). It is also an important regulator of vascular growth coupled with the metabolic and proliferative activities of endothelial cells (Wilhelm et al., 2016). The above experiments confirmed that DT can inhibit tumor angiogenesis in gastric cancer.

The PI3K/Akt/mTOR signaling pathway is not only very important in tumor autophagy and apoptosis (Zhang et al., 2021b; Xiao et al., 2021; Qian et al., 2022), but has also been shown to be involved in the regulation of VEGF, which is closely related to vascular permeability (Junhui et al., 2018). In summary, we speculate that the regulatory effect of DT on tumor angiogenesis in gastric cancer may be related to the regulation of the PI3K/Akt/mTOR pathway. In this study, the effect of DT reduced the expression levels of the PI3K/Akt/mTOR pathway and VEGF. Therefore, we speculated that the regulation effect of DT on the angiogenesis of gastric cancer may be related to the regulation of angiogenesis factor VEGF through the PI3K/Akt/mTOR pathway.

5. Conclusion

In conclusion, this study confirmed that Diterpenoid tanshinones can slow down or stop the development of gastric cancer by inhibiting the proliferation of gastric cancer tumor cells, inducing apoptosis, and inhibiting invasion and metastasis. At the same time, diterpenoid tanshinones were found to be associated with inhibition of tumor angiogenesis. Its mechanism of action may be through the PI3K/Akt/mTOR pathway to affect the expression of angiogenic factor VEGF to achieve the regulation of tumor angiogenesis. This paper provides a new potential approach to treating gastric cancer by simultaneously inhibiting tumor growth and angiogenesis, which has greater theoretical value and clinical reference value.

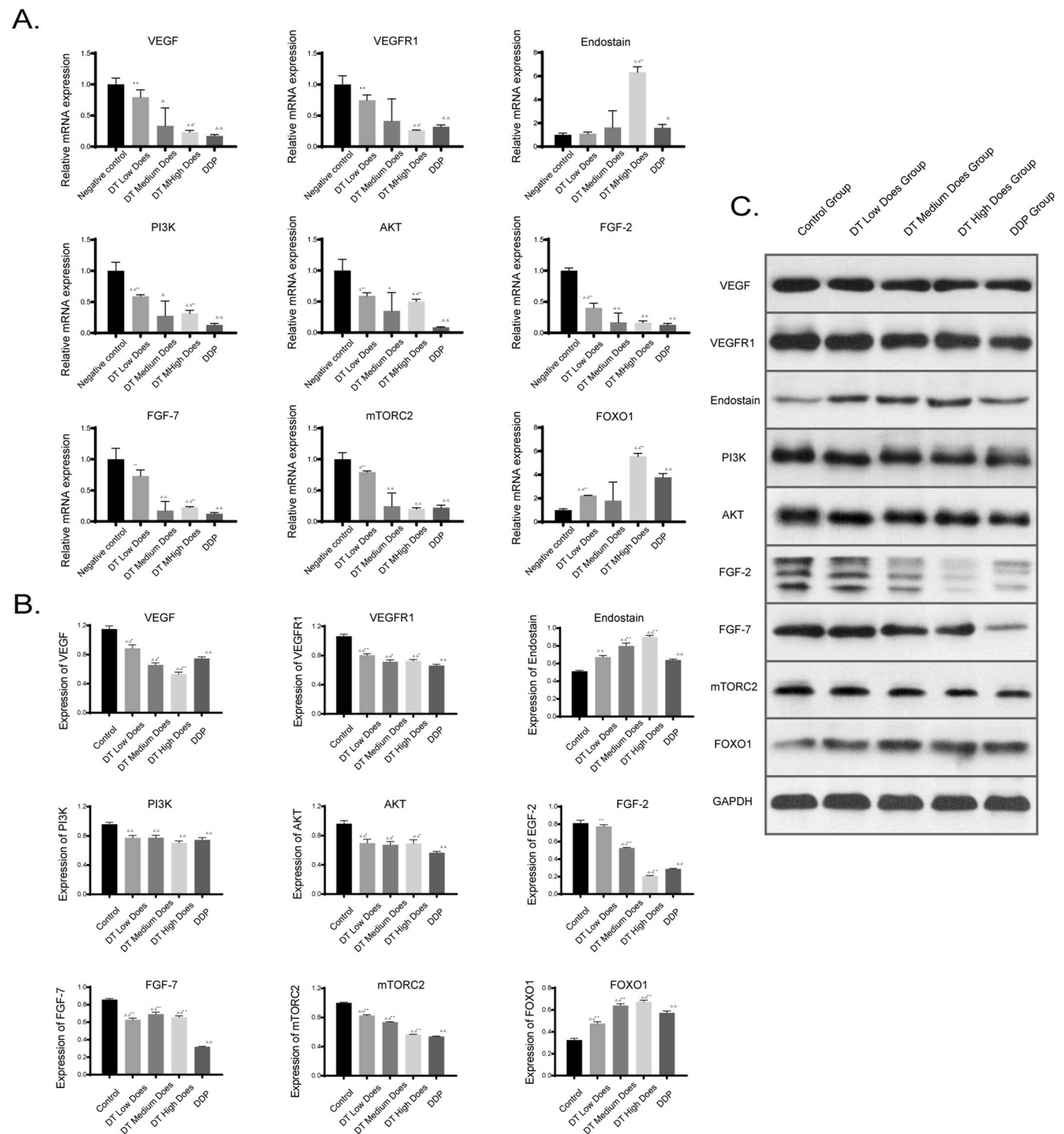


Fig. 5. RT-qPCR and WB experiments of cells (A) RT-qPCR related indicators mRNA expression. Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: * $p < 0.05$; ** $p < 0.01$ (B) Target protein bands in the total protein of cells in each drug intervention group after 24h intervention. Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: * $p < 0.05$; ** $p < 0.01$ (C) Expression of target protein in a total protein of cells in each experimental group (IOD).

CRedit authorship contribution statement

Jie-ru Yu: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yu-yue Liu:** Writing – review & editing, Writing – original draft, Data curation. **Yang-yang Gao:** Writing – original draft, Formal analysis, Data curation. **Li-hui Qian:** Writing – original draft, Methodology,

Investigation. **Jia-lin Qiu:** Writing – original draft. **Pei-pei Wang:** Writing – review & editing, Investigation, Funding acquisition, Data curation. **Guang-ji Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

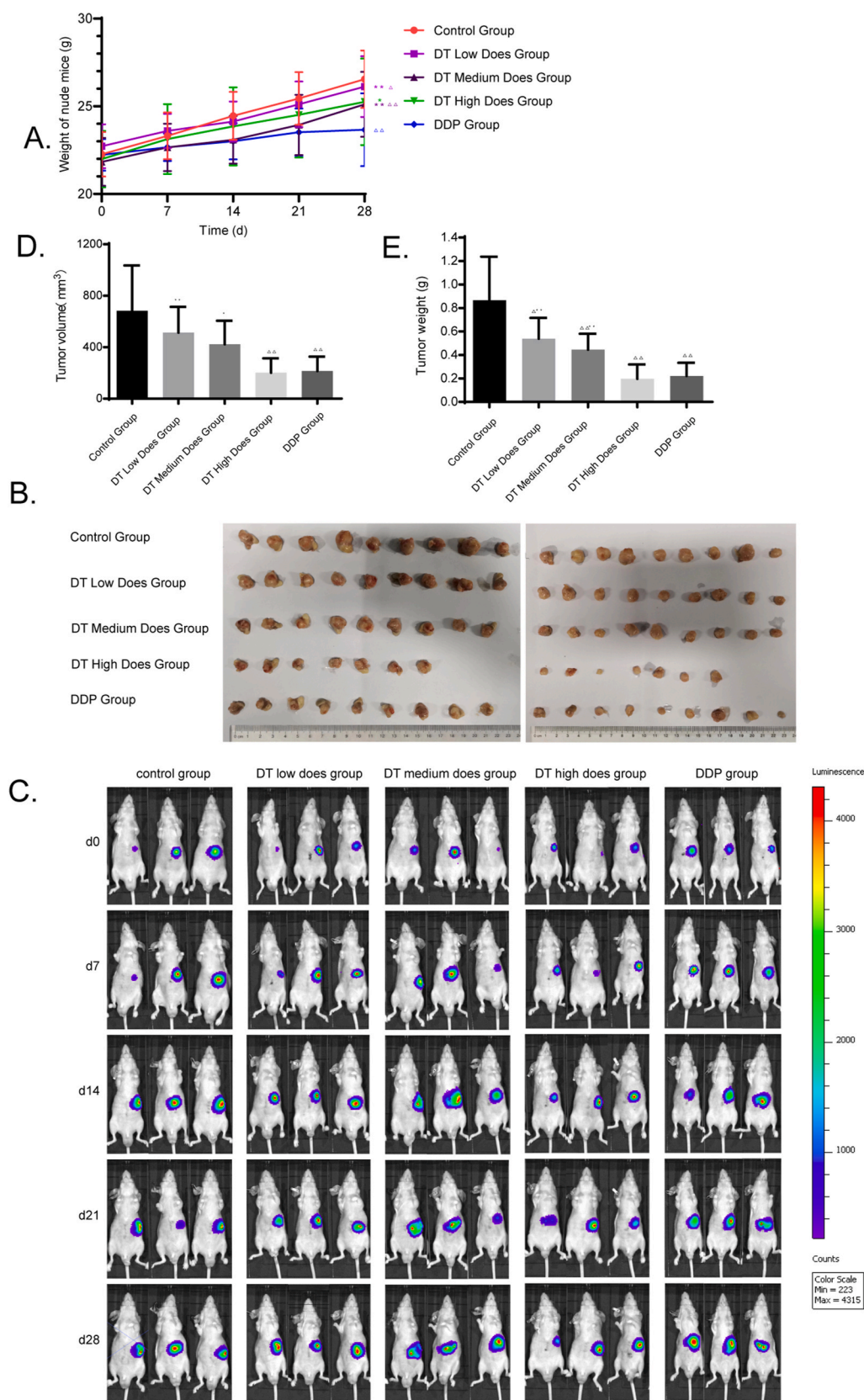


Fig. 6. Body weight and tumor information in nude mice (A) Line chart of body weight of nude mice in each drug intervention group. Note: Compared with the control group: $\Delta p < 0.05$, $\Delta\Delta p < 0.01$; Compared with the cisplatin group: $*p < 0.05$, $**p < 0.01$ (B) Orthotopic transplantation of gastric cancer in nude mice (left: tumor adhered to stomach, right: tumor after peeling) (C) The effect of each drug intervention group on tumor in nude mice at different time (D) Effects of each drug intervention group on tumor volume in nude mice. Note: Compared with the control group: $\Delta p < 0.05$, $\Delta\Delta p < 0.01$; Compared with the cisplatin group: $*p < 0.05$, $**p < 0.01$ (E) Effects of each drug intervention group on tumor weight in nude mice. Note: Compared with the control group: $\Delta p < 0.05$, $\Delta\Delta p < 0.01$; Compared with the cisplatin group: $*p < 0.05$, $**p < 0.01$.

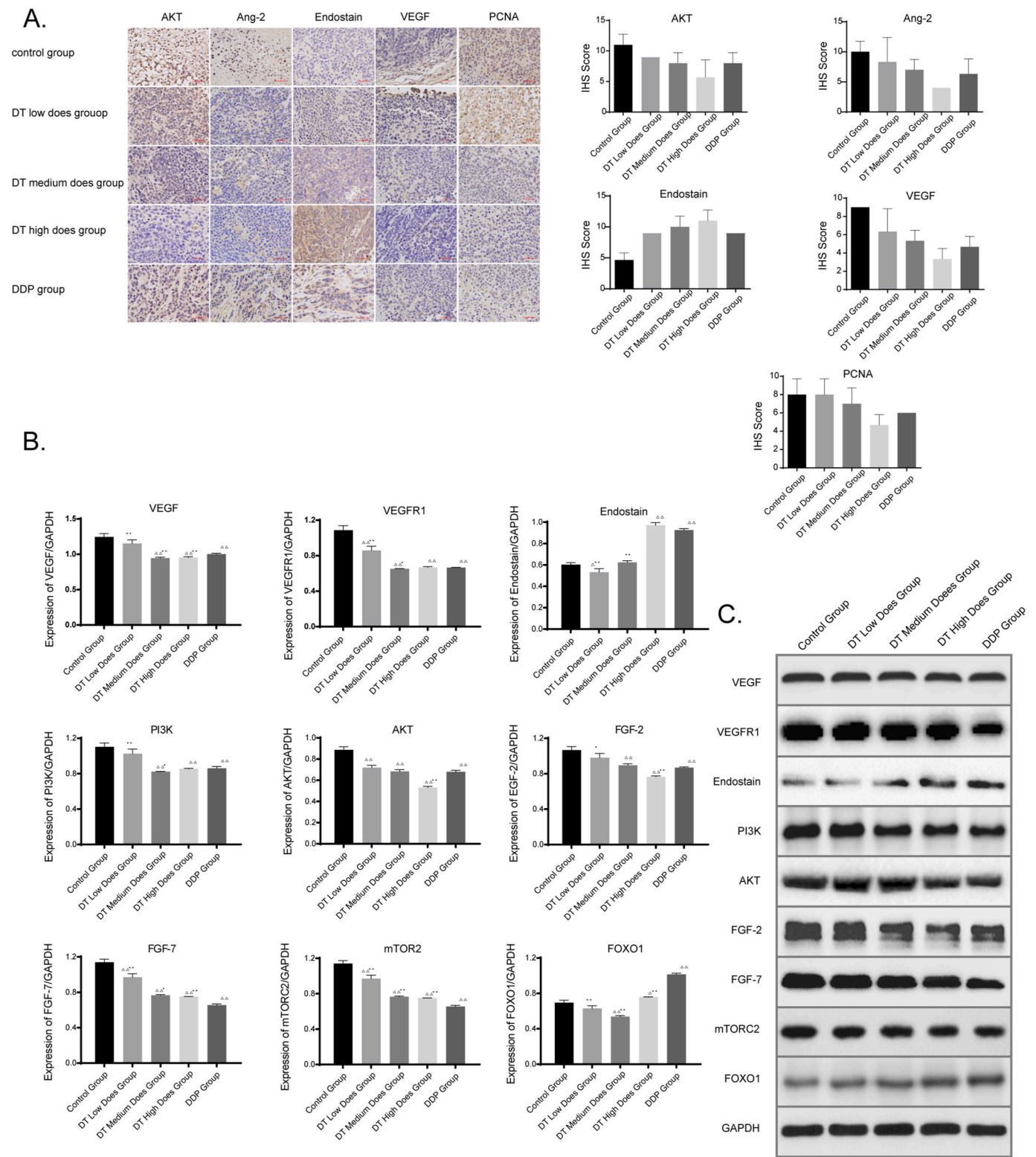


Fig. 7. Immunohistochemistry and WB of tumor in nude mice (A) Immunohistochemical expression in tumor tissue (B) Target protein bands in tumor tissue total protein in each drug intervention group. Note: Compared with the control group: Δ $p < 0.05$, $\Delta\Delta$ $p < 0.01$; Compared with the cisplatin group: * $p < 0.05$; ** $p < 0.01$ (C) Expression of target protein in total protein of tumor tissue in each experimental group (IOD).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2024.117791>.

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Update

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Corrigendum to “Diterpenoid tanshinones inhibit gastric cancer angiogenesis through the PI3K/Akt/mTOR signaling pathway” [J. Ethnopharmacol. 324 (2024) 117791]

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The authors regret < During the final layout process of our manuscript, an inadvertent error occurred in Fig. 7A. Specifically, the image for the DDP group in the VEGF immunohistochemistry results was mistakenly replaced with the image from the DT medium-dose group.

After a carefully check of our experimental record, we found that the error was due to a layout oversight during the finalization of the figures. The corrected Fig. 7 is attached below. The original experimental data and images are accurate, and the error was solely due to a misplacement

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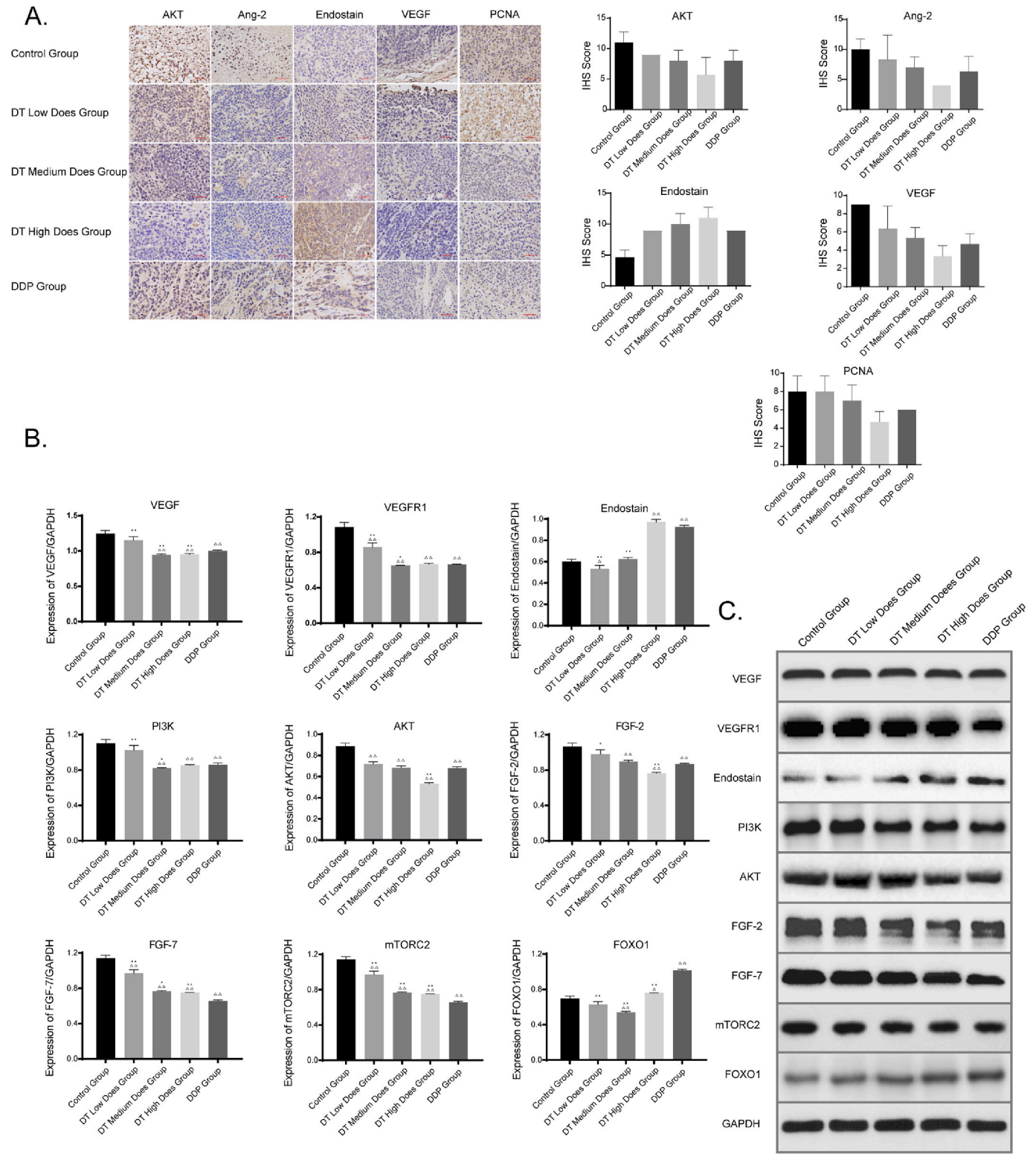
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during figure assembly. >

The authors would like to apologise for any inconvenience caused.
Attachments:



Corrected Fig. 7.

Original experimental data.

